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1

Classified



Geotopia Police Department Violation Record

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Geotopia Police Department Violation Record



2

Name:

John Tripleang

Address:

154 Tangram St.
Geotopia, GT 11223

ID Number:

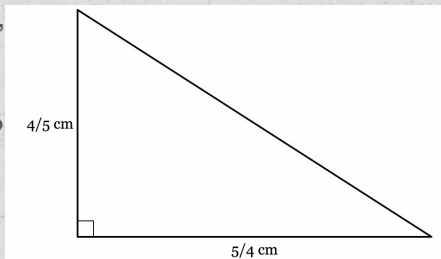
3062471572

Photo:



Details of Violation:

"What is the area, in square centimeters, of the shape below? Express your answer as a fraction in simplest form."



~~John answered:~~

~~Length: $\frac{5}{4}$ cm~~

~~Width: $\frac{4}{5}$ cm~~

~~Area = Length times width~~

~~$$\frac{5}{4} \cdot \frac{4}{5} = \frac{20}{20} = 1 \text{ cm}$$~~

Officer's note:

1) We call the sides of a triangle "base" and "height", not "length and width".

2) Area of a triangle is $\frac{1}{2}bh$. So it should be: $\frac{1}{2} \cdot \frac{5}{4} \cdot \frac{4}{5} = \boxed{\frac{1}{2}}$

3) All area units are with "square". So the answer should be $\boxed{\frac{1}{2} \text{ cm}^2}$



Geotopia Police Department Violation Record



3

Name:
Victoria Tomoro

Address:
39 Triangle Ave.
Geotopia, GT 10066

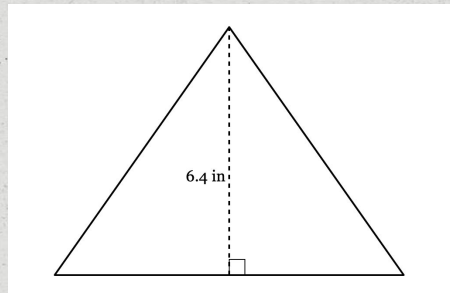
ID Number:
6775503923

Photo:



Details of Violation:

"The area of the triangle below is 28.8 square inches. What is the length of the base?"



~~Victoria's answer:~~

~~$$\frac{28.8}{6.4} = 4.5 \text{ in}^2$$~~

Officer's note:

1) Use area formula to solve:

$$A = \frac{1}{2} \cdot b \cdot h$$

$$28.8 = \frac{1}{2} \cdot 6.4 \cdot h$$

$$28.8 = 3.2 \cdot h$$

Solve for h:

$$\frac{28.8}{3.2} = \frac{3.2h}{3.2}$$

$$9 = h$$

2) Base or height is a "line", not "area" so the unit should be 9 in



Geotopia Police Department Violation Record



4

Name:

Ada Saito

Address:

2699 Trapezoid Ave.
Geotopia, GT 01183

ID Number:

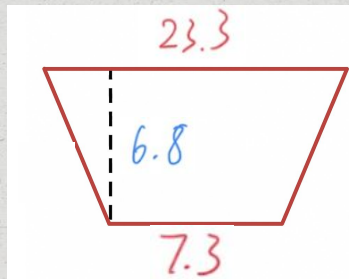
1773811290

Photo:



Details of Violation:

"What is the total area, in square inches, of the shaded sections of the trapezoid below?"



~~Ada's answer:~~

~~Base: 23.3~~

~~Height: 6.8~~

~~Area: $\frac{1}{2}bh$~~

~~$$\frac{1}{2} \cdot (23.3) \cdot (6.8) = 79.22 \text{ in}^2$$~~

Officer's note:

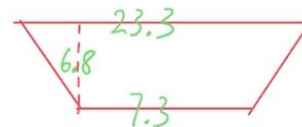
Area of a Trapezoid is

$$A = \frac{1}{2}(b_1 + b_2)h$$

There are 2 bases in a trapezoid. They are always parallel to each other.

base 1: 23.3

base 2: 7.3



$$\text{Area} = \frac{1}{2}(b_1 + b_2)h$$

$$= \frac{1}{2} \cdot (23.3 + 7.3) \cdot 6.8 = 104.04$$

The area is 104.04 in²



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5

Name:
Thomas Vossenber

Address:
86 Tangram St.
Geotopia, GT 11223

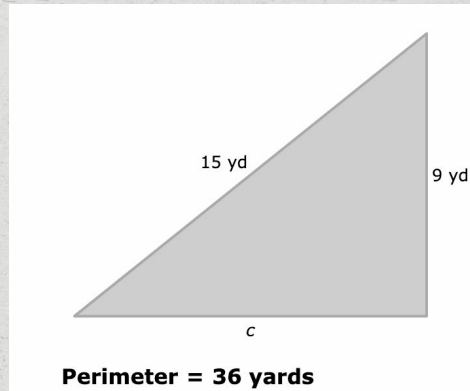
ID Number:
1381077562

Photo:



Details of Violation:

"Given the perimeter of this triangle which is **36** yards, find the value of **c**."



~~Thomas's answer:~~

~~$$\frac{36}{9} = 4 \text{ yd}$$~~

Officer's note:

This is perimeter! Not area!

Add all 3 sides up for the perimeter.

$$15 + 9 + c = 36$$

$$\begin{array}{r} 24 + c = 36 \\ -24 \quad -24 \\ \hline c = 12 \end{array}$$

c is 12 yards long.



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6

Name:
Tyrone Atkins

Address:
525 North GeoPlaza
Geotopia, GT 12355

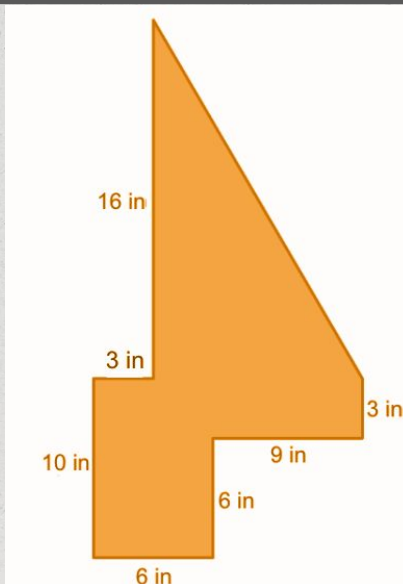
ID Number:
0989952108

Photo:



Details of Violation:

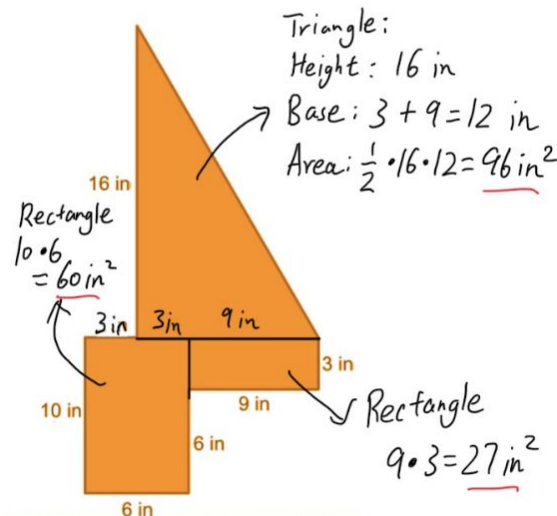
"What is the area
of this polygon?"



~~Tyrone's answer:
 $16 + 2 + 3 + 9 + 6 + 6 + 10$
 $= 52 \text{ in}^2$~~

Officer's note:

We have to break it
down to small polygons
to find the area.



Total Area: $96 + 60 + 27$
 $= 183 \text{ in}^2$



Geotopia Police Department Violation Record



7

Name:
Harry Lopez

Address:
530 South GeoPlaza
Geotopia, GT 12355

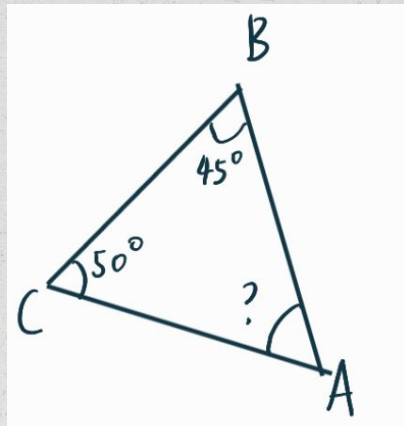
ID Number:
1989954448

Photo:



Details of Violation:

"How do you find the measure of angle A?"



~~$$50 + 45 = 95$$
$$50 - 45 = 5$$~~

Officer's note:

Remember, all 3
angles inside
a triangle always
add up to 180°

$$\begin{aligned}\angle A &= 180^\circ - \angle B - \angle C \\ (\text{Angle A}) &= 180^\circ - 45^\circ - 50^\circ \\ &= \underline{85^\circ}\end{aligned}$$



Geotopia Police Department Violation Record



8

Name:
Jenni Angelo

Address:
1177 GeoTown st
Geotopia, GT 09985

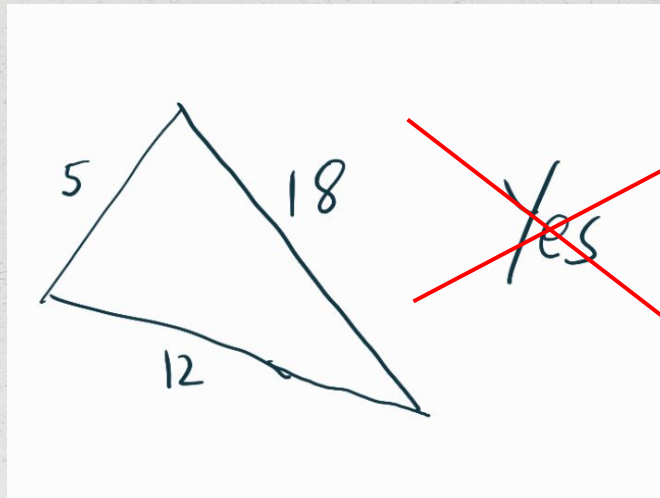
ID Number:
4340433448

Photo:



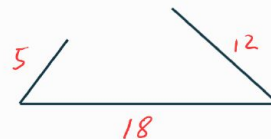
Details of Violation:

"Can 5, 12 and 18 make a triangle?"



Officer's note:

No, 5, 12 and 18 cannot
be the side lengths of
a triangle!



It is because $5+12=17$ which
is less than 18. The sides
of 5 and 12 are not long
enough to reach each other!
Also, $18-12$ is 6 which
is greater than 5. We need
the difference to be less than
the third side.



Geotopia Police Department Violation Record



9

Name:
Jenni Angelo

Address:
1177 GeoTown st
Geotopia, GT 09985

ID Number:
4340433448

Photo:



Details of Violation:

"How many possible lengths of the third side if I have two sides of 5 and 12 to make a triangle?"

~~A lot! $12 - 5 = 7$
So 7 possible lengths!~~

Officer's note:

$$5 + 12 = 17 \quad (\text{Sum of two sides})$$

$$12 - 5 = 7 \quad (\text{Difference of two sides})$$

So, any number between
7 and 17 will work.
(Exclude 7 and 17)

8, 9, 10, 11, 12,
13, 14, 15, 16
Any of these are possible!



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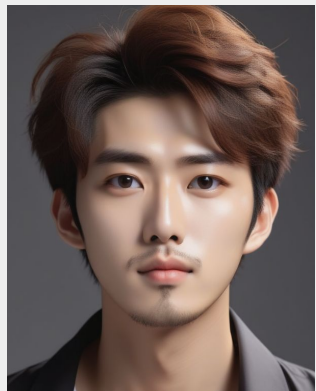
10

Name:
Nelson Wu

Address:
3217 Tangram St.
Geotopia, GT 11223

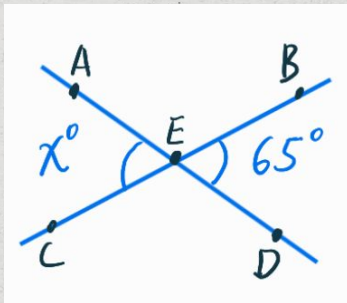
ID Number:
4340111338

Photo:



Details of Violation:

"What is the value of x?"



$$x + 65 = 65x$$

Officer's note:

$\angle AEC$ and $\angle BED$
are **vertical angles**
because they are
opposite to each other and
share the same vertex.
Vertical angles are
EQUAL in measure.
 $x = 65$



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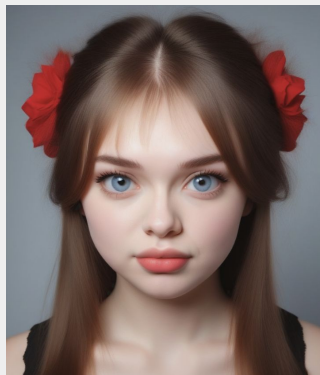
11

Name:
Veronika Siabs

Address:
13 Triangle Ave.
Geotopia, GT 10066

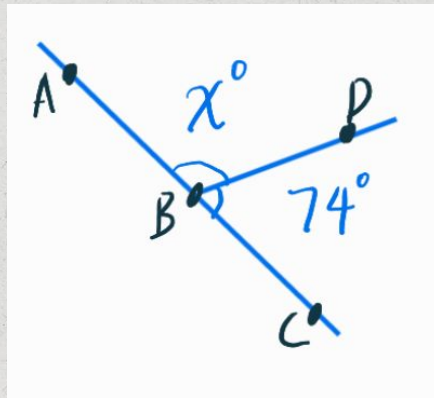
ID Number:
1233633448

Photo:



Details of Violation:

"What is the value of x?"



$$x = 74^\circ$$

Officer's note:

$\angle ABD$ and $\angle DBC$
are *supplementary*
angles because they
add up to 180°
(Measure of a "flat"
angle)

To solve for x :

$$x + 74 = 180$$

$$x = 106$$



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12

Name:
Girogi Madkins

Address:
512 Vertikal Avenue
Geotopia, GT 15440

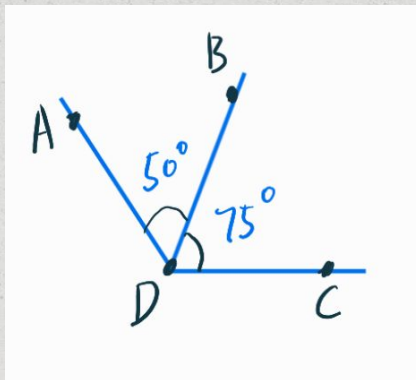
ID Number:
2768122448

Photo:



Details of Violation:

"What is the angle relationship?"



~~$\angle ADB$ and $\angle BDC$
are supplementary
angles.~~

Officer's note:

$\angle ADB$ and $\angle BDC$
are NOT supplementary
because they do not
add up to 180° .
They are adjacent
angles because they
share a side.



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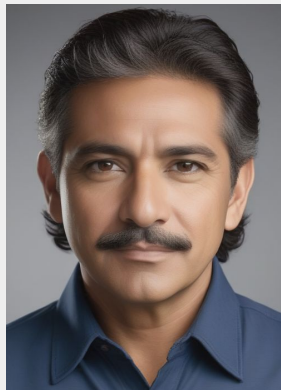
13

Name:
Amani Verez

Address:
18 Supple Avenue
Geotopia, GT 15430

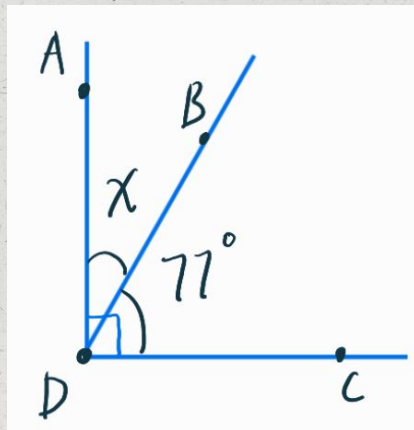
ID Number:
2768129878

Photo:



Details of Violation:

"What is the value of x?"



~~$$x + 77 = 180$$~~

~~$$x = 103$$~~

Officer's note:

$\angle ABD$ and $\angle BDC$
are **Complementary angles**
because they add up
to 90° (right angle)
To solve for x :

$$x + 77 = 90$$

$$x = 90 - 77 = 23$$



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14

Name:
Eugene Johnson

Address:
99 Adjacent street
Geotopia, GT 22440

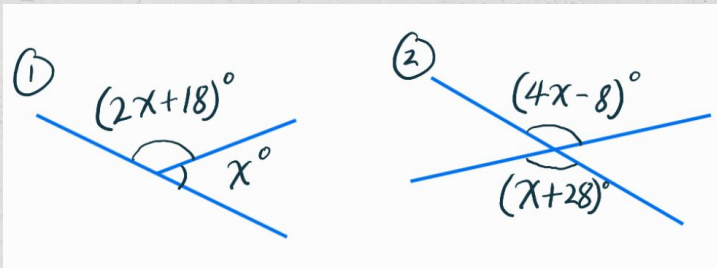
ID Number:
2768333448

Photo:



Details of Violation:

"What is the value of x?"



~~① $2x+18=x$~~
~~② $4x-8+x+28=180$~~

Officer's note:

① Supplementary
 $2x+18+x=180$
 $3x+18=180$
 $3x=162$
 $x=54$

② Vertical
 $4x-8=x+28$
 $4x-x=8+28$
 $3x=36$
 $x=12$



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15

Name:
Alonzo Mbappe

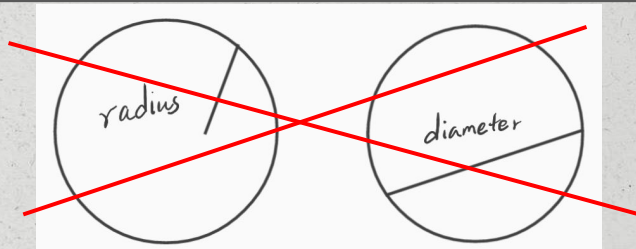
Address:
112 Circle way
Geotopia, GT 22410

ID Number:
1751333448

Photo:

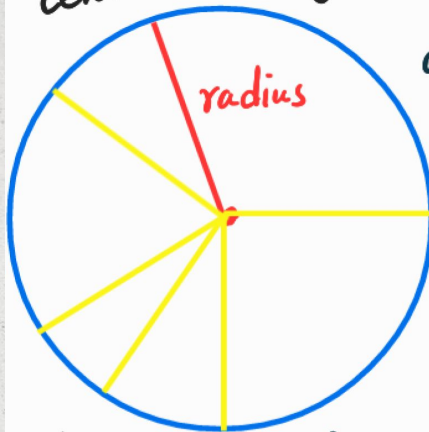


Details of Violation:
"Draw a radius and a
diameter"



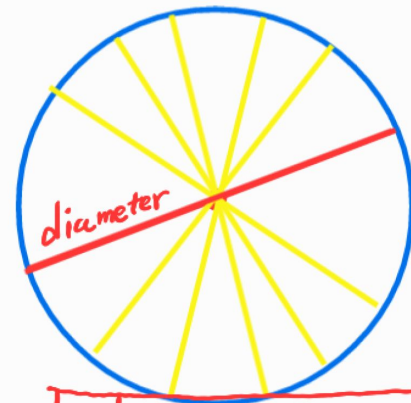
Officer's note:

Radius is from
center to edge



You can draw infinite
amount of radius and diameter

Diameter is from
edge to center to edge



$$d = 2 \cdot r$$

diameter is twice radius



Geotopia Police Department Violation Record



16

Name:
Omar Hassan

Address:
32 Area Road
Geotopia, GT 12410

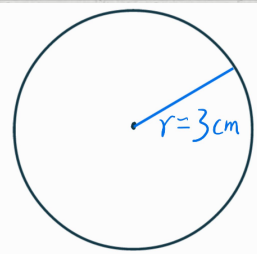
ID Number:
1743883448

Photo:



Details of Violation:

"What is the circumference
if the radius is 3cm?"



~~$$3 \cdot \pi = 3\pi \text{ cm}$$
$$3 \cdot 3.14 = 9.42 \text{ cm}$$~~

Officer's note:

If you know the radius,
use $C = 2\pi r$

Circumference is double radius
times π

$$C = 2 \cdot \pi \cdot 3$$
$$= \underline{6\pi} \quad = 6 \cdot 3.14 = \underline{18.84}$$

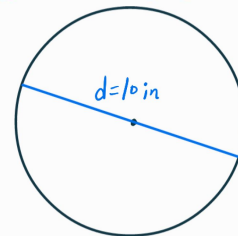
$6\pi \text{ cm}$ is in terms of π
 18.84 cm is in decimal

If you know the diameter,
use $C = \pi d$

Circumference is π times diameter.

$$C = \pi \cdot 10 = \underline{10\pi \text{ in}} = \underline{10 \cdot 3.14} = \underline{31.4 \text{ in}}$$

(in terms of π) (in decimal)





Geotopia Police Department Violation Record



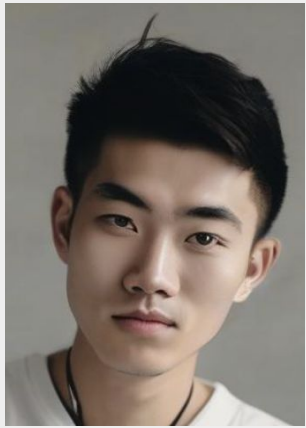
17

Name:
Hongqing Wei

Address:
119 Tianan Avenue
Geotopia, GT 11310

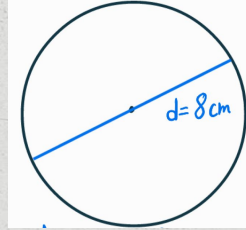
ID Number:
0743903440

Photo:



Details of Violation:

"What is the area if the diameter is 8cm?"



~~$$\begin{aligned} \text{Area} &= 8^2 \cdot \pi \\ &= 64\pi \text{ cm} \end{aligned}$$~~

Officer's note:

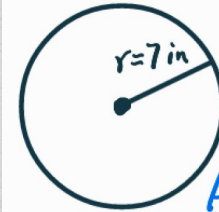
We always need radius to solve for area!

$$A = \pi r^2$$

Area is π times radius square,
 $d = 8\text{cm} \rightarrow r = 4\text{cm}$ (half)

$$\begin{aligned} A &= \pi \cdot 4^2 = 16\pi \text{ cm}^2 \\ &= 16 \cdot 3.14 \\ &= 50.24 \text{ cm}^2 \end{aligned}$$

(16π is in terms of π . 50.24 is in decimal.)



We have the radius 7 in.

$$\begin{aligned} A &= \pi \cdot 7^2 = 49\pi \\ &= 49 \cdot 3.14 \\ &= 153.86 \end{aligned}$$

Either $49\pi \text{ in}^2$
or 153.86 in^2



Geotopia Police Department Violation Record



18

Name:
Yuki Nakamura

Address:
119 Tianan Avenue
Geotopia, GT 11310

ID Number:
0743903441

Photo:

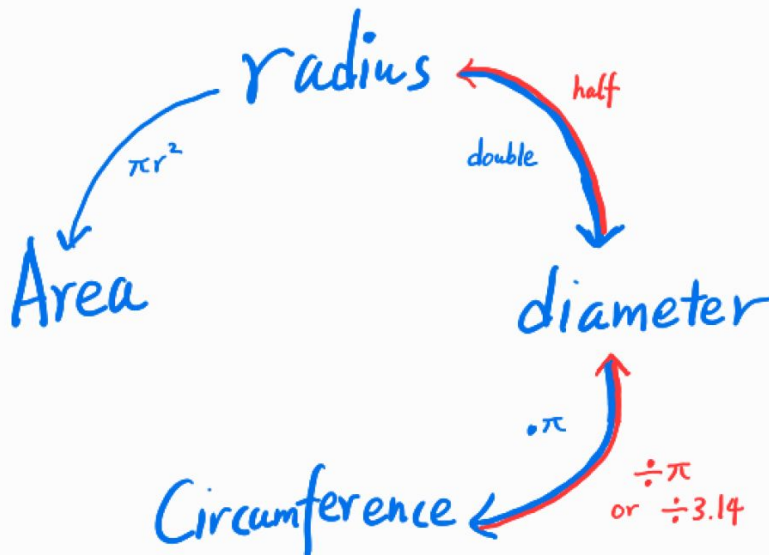


Details of Violation:

"The circumference of a circle is 28.26 cm. What is the area?"

~~$$C = 28.26 \text{ cm}$$
$$d = 28.26 \div \pi = 9 \text{ cm}$$
$$A = 9^2 \cdot \pi = 254.34 \text{ cm}^2$$~~

Officer's note:



Follow this chart.
If we have $C = 28.26$,
then $\div 3.14$ is 9 for diameter.
 $9 \div 2 = 4.5$ for radius.
 $4.5^2 \cdot \pi = \underline{\underline{63.59 \text{ cm}^2}}$



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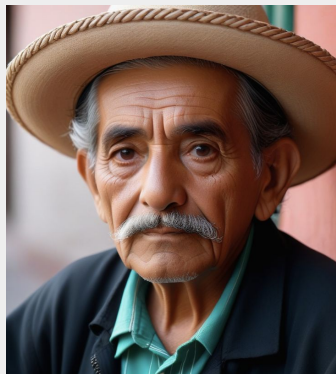
19

Name:
Alejandro García

Address:
89 Volume Avenue
Geotopia, GT 12341

ID Number:
0678833422

Photo:

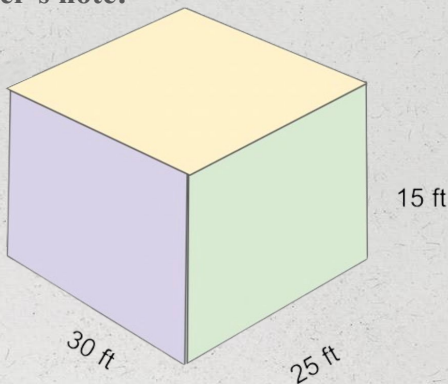


Details of Violation:

"What is the volume of the rectangular prism with 30 ft, 25 ft, and 15 ft as the length, width and height?"

~~Volume?
 $30 + 25 + 15 = 70 \text{ ft}^3$~~

Officer's note:



Volume:

Length * Width * Height

30

*

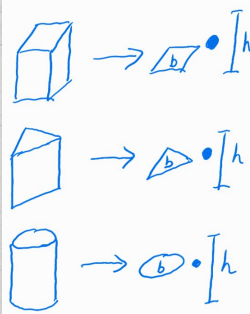
25

*

15

= 11250 ft³

Volume of a prism is always
the base area times the height.





Geotopia Police Department Violation Record



20

Name:
Aahana Gupta

Address:
47 Surface Avenue
Geotopia, GT 10011

ID Number:
1238567723

Photo:



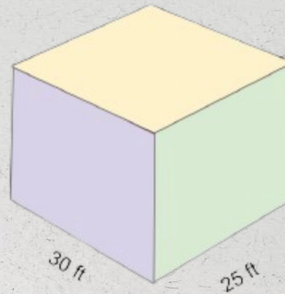
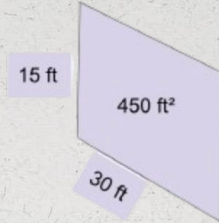
Details of Violation:

"What is the surface area of this rectangular prism which has length, width and height in 30 ft, 25 ft and 15 ft?"

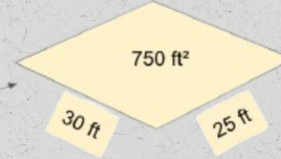
~~Surface Area?~~
 ~~$30 \times 25 \times 15 = 11250 \text{ ft}^2$~~

Officer's note:

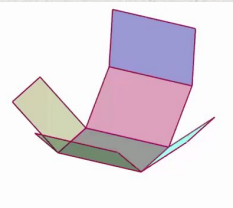
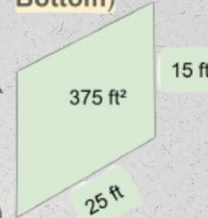
This face is the Front
(The area is the same as the Back)



This face is the Top
(The area is the same as the Bottom)



This face is the right
(The area is the same as the left)



Surface Area:

Top	$750 * 2$
Bottom	+
Left	$375 * 2$
Right	+
Front	$450 * 2$
Back	

Sum	3150	ft ²
------------	------	-----------------



Geotopia Police Department Violation Record



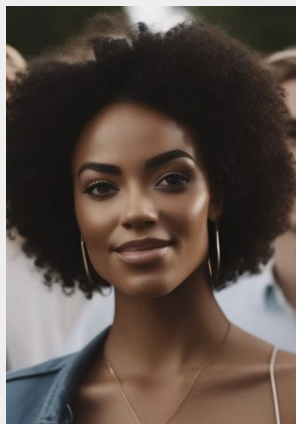
21

Name:
Stephenie Anderson

Address:
135 Prism Avenue
Geotopia, GT 10323

ID Number:
1456119973

Photo:

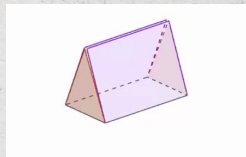
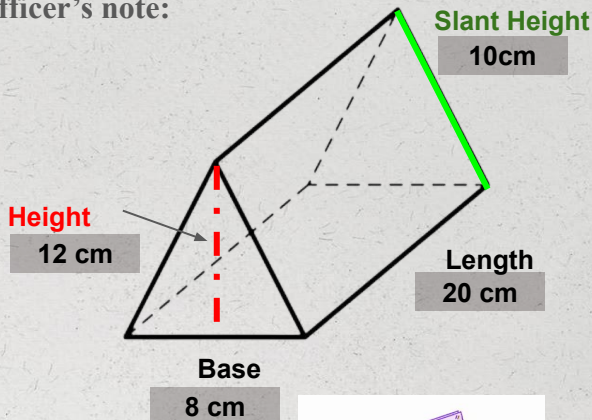


Details of Violation:

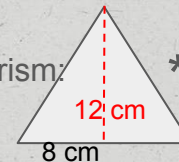
"What is the **surface area** and **volume** of a **triangular prism** which has base of 8 cm, height of 12 cm, length of 20 cm and slant height of 10 cm?"

~~Volume: $12 \times 8 \times 20 \times 10 = 19200 \text{ ft}^3$~~
~~Surface Area: $12 \times 8 \times 20 = 1920 \text{ ft}^2$~~

Officer's note:



**Volume of
Triangular Prism:**



*

Length = 20 cm

$$(\frac{1}{2} \times 8 \times 12) \times 20 = 960 \text{ cm}^3$$

Surface Area:

Front + Back + Right + Left + Bottom

The diagram shows the components of the surface area calculation: two triangles (front and back) and three rectangles (right, left, and bottom). The triangles have a base of 8 cm and a height of 12 cm. The rectangles have dimensions 8 cm by 20 cm, 10 cm by 20 cm, and 10 cm by 20 cm.

$$\frac{1}{2} \times 8 \times 12 + \frac{1}{2} \times 8 \times 12 + 20 \times 10 + 20 \times 10 + 8 \times 20 = 960 \text{ cm}^2$$

$$\frac{1}{2} \times 8 \times 12 + \frac{1}{2} \times 8 \times 12 + 20 \times 10 + 20 \times 10 + 8 \times 20$$



Geotopia Police Department Violation Record



22

Name:
Acozy Guy

Address:
3089 Pyramid Street
Geotopia, GT 28009

ID Number:
1347890445

Photo:

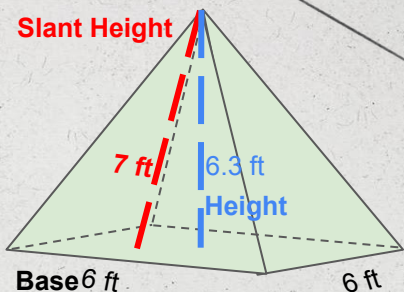
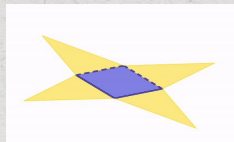


Details of Violation:

"What is the **surface area** and **volume** of a square pyramid which has base of 6 ft, height of 6.3 ft, slant height of 7 ft?"

~~Volume: $7 \times 6 \times 6 = 252 \text{ ft}^3$
Surface Area: $21 \times 4 = 84 \text{ ft}^2$~~

Officer's note:



Volume is the $\frac{1}{3}$ the base triangle times the height

Area of front triangle = $\frac{1}{2} \times 7 \times 6 = 21 \text{ ft}^2$

Volume = $\frac{1}{3} \times 21 \times 6.3 = 44.1 \text{ ft}^3$

Area of front triangle = $\frac{1}{2} \times 7 \times 6 = 21 \text{ ft}^2$

4 congruent triangles: $21 \text{ ft}^2 \times 4 = 84 \text{ ft}^2$

Base = $6 \times 6 = 36 \text{ ft}^2$ (this is a square)

Surface Area = $36 + 84 = 120 \text{ ft}^2$



Geotopia Police Department Violation Record



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Name:
Crystal Liam

Address:
88 Cross Pl
Geotopia, GT 10086

ID Number:
5163036819

Photo:

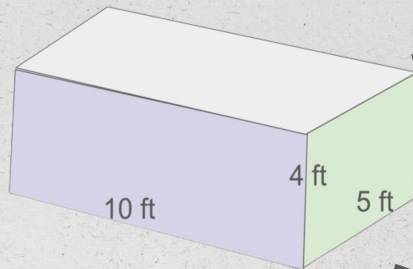


Details of Violation:

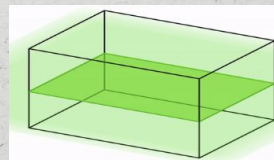
"What is the area of the **cross sections**?"

~~Cross Section is
crossing a section,
right?~~

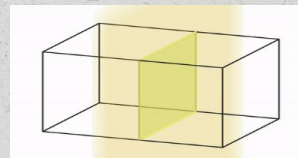
Officer's note:



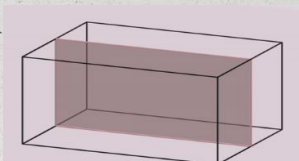
The cross section of this prism
is a rectangle.



If we cut horizontally:
 $10 * 5 = 50 \text{ ft}^2$



If we cut vertically:
 $4 * 5 = 20 \text{ ft}^2$



If we cut vertically from the right:
 $10 * 4 = 40 \text{ ft}^2$